job_8-Evaluation report

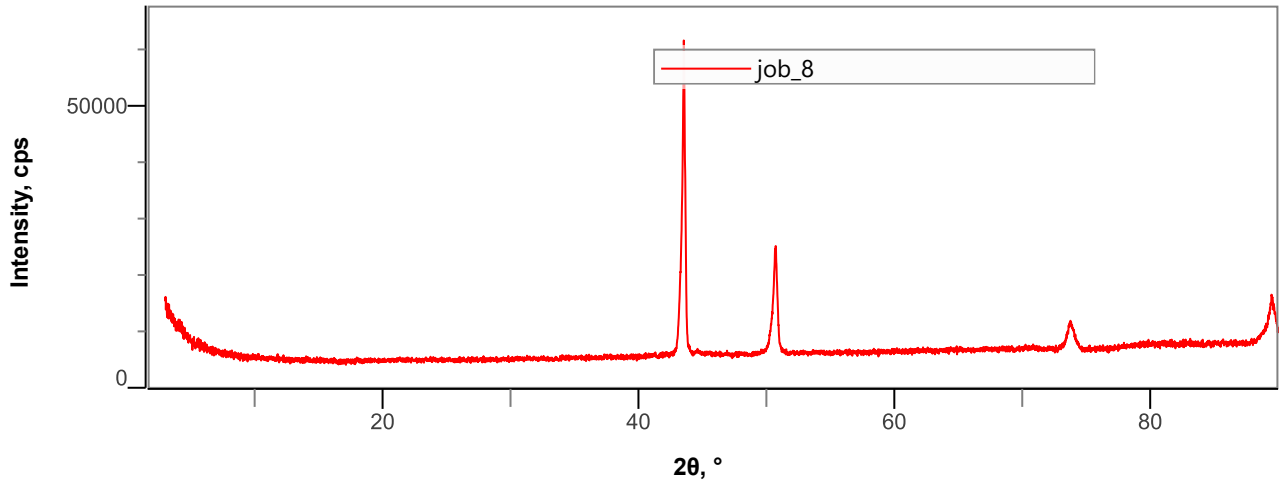
General information

Analysis date	2025-12-17 11:00:17	Measurement start time	2025-11-23 10:35:02
Analyst	Administrator	Operator	Administrator
Sample name	job_8	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\job_8.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	ASC-8	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

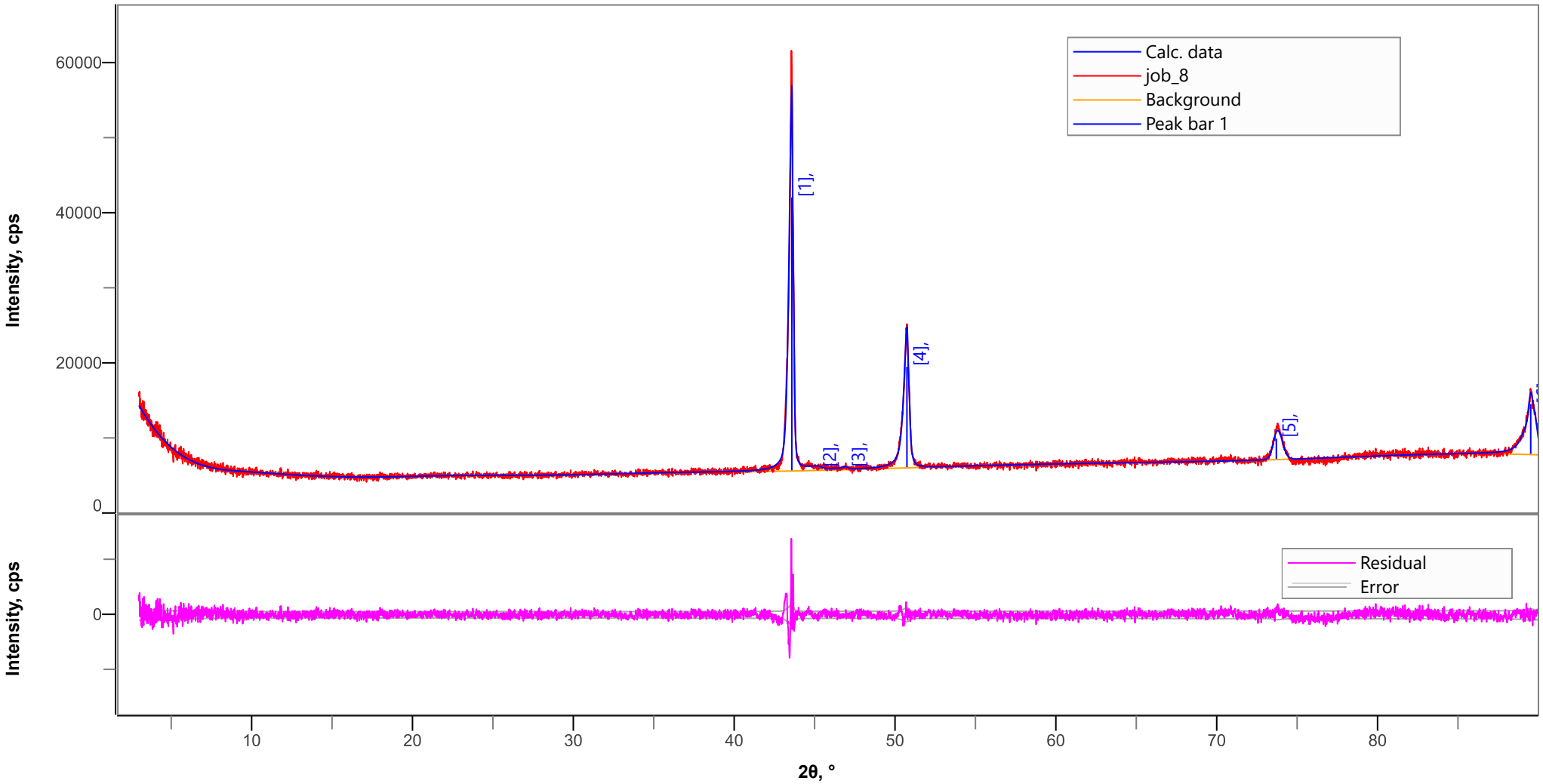
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	3.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

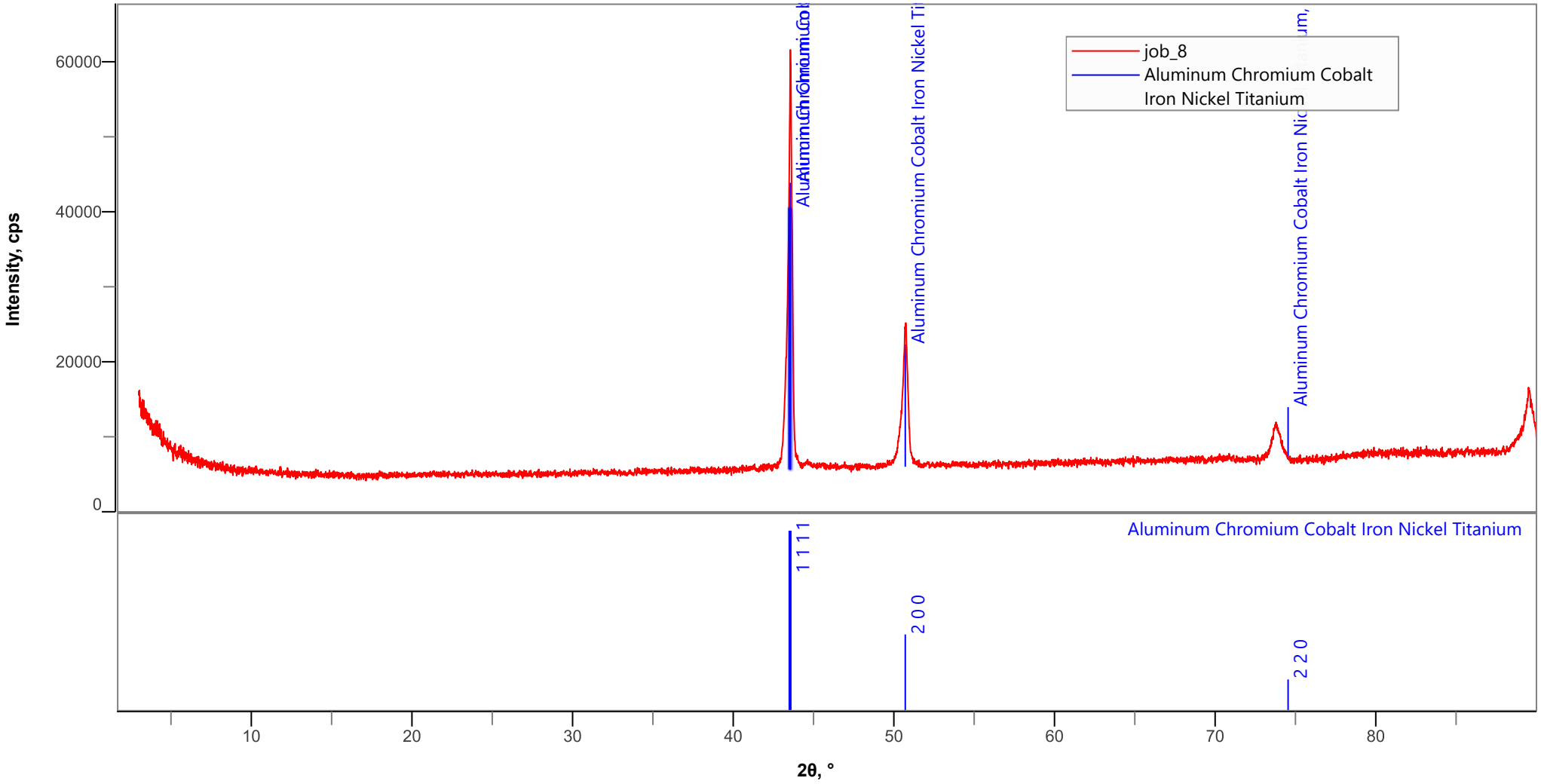
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
Aluminum Chromium Cobalt Iron Nickel Titanium	Ti0.092 Cr0.181 Fe0.312 Co0.187 Ni0.200 Al0.028	0.472	S/M:PDF-5+ 2026	225 : Fm-3m	04-028-5026

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, cps	FWHM, °	Int. I., cps°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay (ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	43.5615 (18)	2.075796 (8)	36379 (600)	0.2754 (19)	12802(74)	0.352(8)	3.84 (17)	0.428 (14)	0.81(5)	324(2)	Aluminum Chromium Cobalt Iron Nickel Titanium: 1 1 1	Ti0.092 Cr0.181 Fe0.312 Co0.187 Ni0.200 Al0.028	04-028-5026	100.00	Split Pseudo-Voigt	-	-	-	-	-	
2	45.12 (5)	2.008 (2)	329 (17)	1.5 (3)	637 (106)	1.9 (4)	3.84 (17)	0.428 (14)	0.81(5)	59(11)	Unknown			4.98	Split Pseudo-Voigt	-	-	-	-	-	
3	46.91 (5)	1.9354 (19)	251 (13)	0.5 (3)	266 (69)	1.1 (3)	5(9)	1.5 (4)	1.5(9)	181 (102)	Unknown			2.08	Split Pseudo-Voigt	-	-	-	-	-	
4	50.714 (5)	1.798667 (16)	13437 (344)	0.335 (5)	6371 (43)	0.474 (15)	2.8 (2)	0.71 (3)	0.91(8)	274(4)	Aluminum Chromium Cobalt Iron Nickel Titanium: 2 0 0	Ti0.092 Cr0.181 Fe0.312 Co0.187 Ni0.200 Al0.028	04-028-5026	49.77	Split Pseudo-Voigt	-	-	-	-	-	
5	73.667 (5)	1.284992 (8)	2788 (108)	0.612 (15)	1851 (57)	0.66 (5)	0.78 (10)	0.13 (13)	0.00(12)	169(4)	Unknown			14.46	Split Pseudo-Voigt	-	-	-	-	-	
6	89.475 (8)	1.094399 (7)	6715 (214)	0.440 (15)	5153 (59)	0.77 (3)	1.28 (12)	1.48 (5)	0.69(11)	266(9)	Unknown			40.25	Split Pseudo-Voigt	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
Aluminum Chromium Cobalt Iron Nickel Titanium	3.59936	3.59936	3.59936	90.000	90.000	90.000

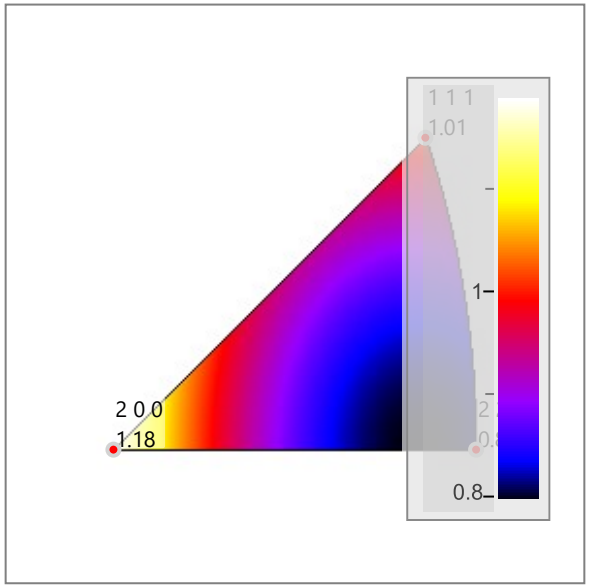
d-I List

Aluminum Chromium Cobalt Iron Nickel Titanium

No.	2θ, °	d, Å	h k l	Norm. I.
1	43.51447	2.07809	1 1 1	100.00
2	50.68373	1.79968	2 0 0	45.08
3	74.50220	1.27257	2 2 0	22.70

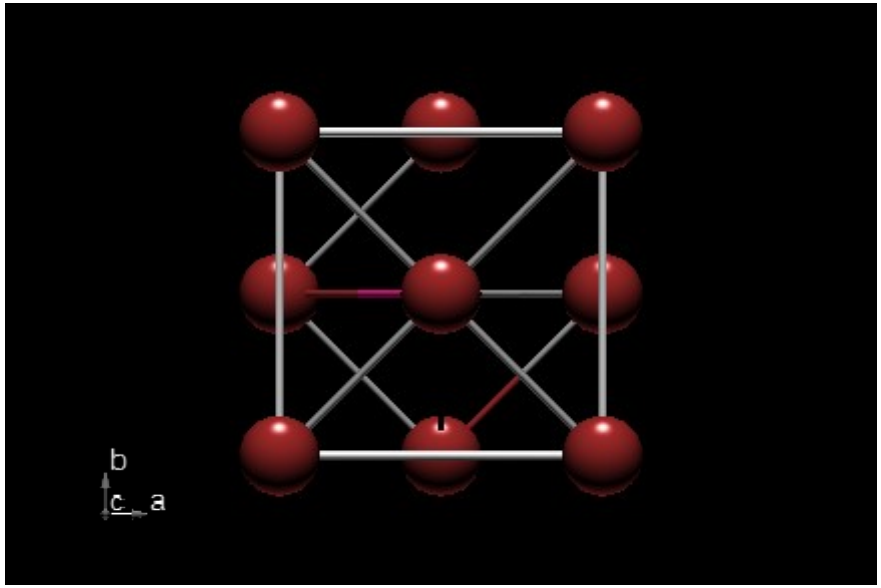
Simple Inverse Pole Figure

Aluminum Chromium Cobalt Iron Nickel Titanium



Crystal Structure

Aluminum Chromium Cobalt Iron Nickel Titanium



Structure parameters

Element	x	y	z	Occ.	B	Charge	Multiplicity	Valence	Bond Valence Sum
Fe1(Fe)	0.000000	0.000000	0.000000	0.312	0.500	0	4	0.000	0.000
Ni1(Ni)	0.000000	0.000000	0.000000	0.200	0.500	0	4	0.000	0.000
Co1(Co)	0.000000	0.000000	0.000000	0.187	0.500	0	4	0.000	0.000
Cr1(Cr)	0.000000	0.000000	0.000000	0.181	0.500	0	4	0.000	0.000
Ti1(Ti)	0.000000	0.000000	0.000000	0.092	0.500	0	4	0.000	0.000
Al1(Al)	0.000000	0.000000	0.000000	0.028	0.500	0	4	0.000	0.000

Distance

Bond Def.	Distance, Å
Ni1-Co1(1/2+x,y,-1/2+z)	2.54513
Ni1-Co1(-1/2+x,y,1/2+z)	2.54513
Ni1-Co1(x,1/2+y,1/2+z)	2.54513
Ni1-Co1(1/2+x,y,1/2+z)	2.54513
Ni1-Co1(1/2+x,-1/2+y,z)	2.54513
Ni1-Co1(-1/2+x,1/2+y,z)	2.54513
Ni1-Co1(1/2+x,1/2+y,z)	2.54513
Ni1-Co1(x,1/2+y,-1/2+z)	2.54513
Ni1-Co1(x,-1/2+y,1/2+z)	2.54513
Ni1-Cr1(1/2+x,1/2+y,z)	2.54513
Ni1-Cr1(-1/2+x,1/2+y,z)	2.54513
Ni1-Cr1(1/2+x,-1/2+y,z)	2.54513
Ni1-Cr1(1/2+x,y,1/2+z)	2.54513
Ni1-Cr1(-1/2+x,y,1/2+z)	2.54513
Ni1-Cr1(1/2+x,y,-1/2+z)	2.54513
Ni1-Cr1(x,1/2+y,1/2+z)	2.54513
Ni1-Cr1(x,-1/2+y,1/2+z)	2.54513
Ni1-Cr1(x,1/2+y,-1/2+z)	2.54513
Ni1-Al1(1/2+x,-1/2+y,z)	2.54513
Ni1-Al1(1/2+x,y,1/2+z)	2.54513
Ni1-Al1(-1/2+x,y,1/2+z)	2.54513
Ni1-Al1(1/2+x,y,-1/2+z)	2.54513
Ni1-Al1(x,1/2+y,1/2+z)	2.54513
Ni1-Al1(x,-1/2+y,1/2+z)	2.54513
Ni1-Al1(x,1/2+y,-1/2+z)	2.54513
Ni1-Al1(1/2+x,1/2+y,z)	2.54513

Distance

Bond Def.	Distance, Å
Ni1-Al1(-1/2+x,1/2+y,z)	2.54513
Co1-Co1(x,-1/2+y,1/2+z)	2.54513
Co1-Co1(x,1/2+y,-1/2+z)	2.54513
Co1-Co1(1/2+x,1/2+y,z)	2.54513
Co1-Co1(x,1/2+y,1/2+z)	2.54513
Co1-Co1(1/2+x,-1/2+y,z)	2.54513
Co1-Co1(1/2+x,y,1/2+z)	2.54513
Co1-Co1(-1/2+x,y,1/2+z)	2.54513
Co1-Co1(-1/2+x,1/2+y,z)	2.54513
Co1-Co1(1/2+x,y,-1/2+z)	2.54513
Co1-Cr1(1/2+x,y,-1/2+z)	2.54513
Co1-Cr1(x,-1/2+y,1/2+z)	2.54513
Co1-Cr1(x,1/2+y,-1/2+z)	2.54513
Co1-Cr1(1/2+x,1/2+y,z)	2.54513
Co1-Cr1(-1/2+x,1/2+y,z)	2.54513
Co1-Cr1(1/2+x,-1/2+y,z)	2.54513
Co1-Cr1(1/2+x,y,1/2+z)	2.54513
Co1-Cr1(-1/2+x,y,1/2+z)	2.54513
Co1-Cr1(x,1/2+y,1/2+z)	2.54513
Co1-Ti1(-1/2+x,1/2+y,z)	2.54513
Co1-Ti1(1/2+x,1/2+y,z)	2.54513
Co1-Ti1(x,1/2+y,-1/2+z)	2.54513
Co1-Ti1(x,-1/2+y,1/2+z)	2.54513
Co1-Ti1(x,1/2+y,1/2+z)	2.54513
Co1-Ti1(1/2+x,y,-1/2+z)	2.54513
Co1-Ti1(-1/2+x,y,1/2+z)	2.54513

Distance

Bond Def.	Distance, Å
Co1-Ti1(1/2+x,y,1/2+z)	2.54513
Co1-Ti1(1/2+x,-1/2+y,z)	2.54513
Co1-Al1(x,1/2+y,1/2+z)	2.54513
Co1-Al1(x,1/2+y,-1/2+z)	2.54513
Co1-Al1(1/2+x,1/2+y,z)	2.54513
Co1-Al1(-1/2+x,1/2+y,z)	2.54513
Co1-Al1(x,-1/2+y,1/2+z)	2.54513
Co1-Al1(1/2+x,-1/2+y,z)	2.54513
Co1-Al1(1/2+x,y,1/2+z)	2.54513
Co1-Al1(-1/2+x,y,1/2+z)	2.54513
Co1-Al1(1/2+x,y,-1/2+z)	2.54513
Fe1-Ni1(1/2+x,y,-1/2+z)	2.54513
Fe1-Ni1(-1/2+x,y,1/2+z)	2.54513
Fe1-Ni1(1/2+x,y,1/2+z)	2.54513
Fe1-Ni1(1/2+x,-1/2+y,z)	2.54513
Fe1-Ni1(-1/2+x,1/2+y,z)	2.54513
Fe1-Ni1(1/2+x,1/2+y,z)	2.54513
Fe1-Ni1(x,1/2+y,-1/2+z)	2.54513
Fe1-Ni1(x,1/2+y,1/2+z)	2.54513
Fe1-Ni1(x,-1/2+y,1/2+z)	2.54513
Fe1-Co1(1/2+x,y,-1/2+z)	2.54513
Fe1-Co1(-1/2+x,y,1/2+z)	2.54513
Fe1-Co1(1/2+x,y,1/2+z)	2.54513
Fe1-Co1(1/2+x,1/2+y,z)	2.54513
Fe1-Co1(1/2+x,-1/2+y,z)	2.54513
Fe1-Co1(-1/2+x,1/2+y,z)	2.54513

Distance

Bond Def.	Distance, Å
Fe1-Co1(x,1/2+y,-1/2+z)	2.54513
Fe1-Co1(x,-1/2+y,1/2+z)	2.54513
Fe1-Co1(x,1/2+y,1/2+z)	2.54513
Fe1-Fe1(1/2+x,-1/2+y,z)	2.54513
Fe1-Fe1(1/2+x,y,1/2+z)	2.54513
Fe1-Fe1(-1/2+x,y,1/2+z)	2.54513
Fe1-Fe1(1/2+x,y,-1/2+z)	2.54513
Fe1-Fe1(x,1/2+y,1/2+z)	2.54513
Fe1-Fe1(x,-1/2+y,1/2+z)	2.54513
Fe1-Fe1(x,1/2+y,-1/2+z)	2.54513
Fe1-Fe1(1/2+x,1/2+y,z)	2.54513
Fe1-Fe1(-1/2+x,1/2+y,z)	2.54513
Fe1-Cr1(x,1/2+y,-1/2+z)	2.54513
Fe1-Cr1(1/2+x,-1/2+y,z)	2.54513
Fe1-Cr1(-1/2+x,1/2+y,z)	2.54513
Fe1-Cr1(1/2+x,1/2+y,z)	2.54513
Fe1-Cr1(x,-1/2+y,1/2+z)	2.54513
Fe1-Cr1(x,1/2+y,1/2+z)	2.54513
Fe1-Cr1(1/2+x,y,-1/2+z)	2.54513
Fe1-Cr1(-1/2+x,y,1/2+z)	2.54513
Fe1-Cr1(1/2+x,y,1/2+z)	2.54513
Fe1-Ti1(1/2+x,y,-1/2+z)	2.54513
Fe1-Ti1(-1/2+x,y,1/2+z)	2.54513
Fe1-Ti1(1/2+x,y,1/2+z)	2.54513
Fe1-Ti1(1/2+x,-1/2+y,z)	2.54513
Fe1-Ti1(-1/2+x,1/2+y,z)	2.54513

Distance

Bond Def.	Distance, Å
Fe1-Ti1(1/2+x,1/2+y,z)	2.54513
Fe1-Ti1(x,1/2+y,-1/2+z)	2.54513
Fe1-Ti1(x,-1/2+y,1/2+z)	2.54513
Fe1-Ti1(x,1/2+y,1/2+z)	2.54513
Fe1-Al1(1/2+x,1/2+y,z)	2.54513
Fe1-Al1(x,1/2+y,-1/2+z)	2.54513
Fe1-Al1(x,-1/2+y,1/2+z)	2.54513
Fe1-Al1(x,1/2+y,1/2+z)	2.54513
Fe1-Al1(-1/2+x,1/2+y,z)	2.54513
Fe1-Al1(1/2+x,y,-1/2+z)	2.54513
Fe1-Al1(-1/2+x,y,1/2+z)	2.54513
Fe1-Al1(1/2+x,y,1/2+z)	2.54513
Fe1-Al1(1/2+x,-1/2+y,z)	2.54513
Cr1-Cr1(x,1/2+y,1/2+z)	2.54513
Cr1-Cr1(x,-1/2+y,1/2+z)	2.54513
Cr1-Cr1(x,1/2+y,-1/2+z)	2.54513
Cr1-Cr1(1/2+x,1/2+y,z)	2.54513
Cr1-Cr1(-1/2+x,1/2+y,z)	2.54513
Cr1-Cr1(1/2+x,-1/2+y,z)	2.54513
Cr1-Cr1(1/2+x,y,1/2+z)	2.54513
Cr1-Cr1(-1/2+x,y,1/2+z)	2.54513
Cr1-Cr1(1/2+x,y,-1/2+z)	2.54513
Cr1-Ti1(x,1/2+y,1/2+z)	2.54513
Cr1-Ti1(x,-1/2+y,1/2+z)	2.54513
Cr1-Ti1(x,1/2+y,-1/2+z)	2.54513
Cr1-Ti1(1/2+x,1/2+y,z)	2.54513

Distance

Bond Def.	Distance, Å
Cr1-Ti1(-1/2+x,1/2+y,z)	2.54513
Cr1-Ti1(1/2+x,-1/2+y,z)	2.54513
Cr1-Ti1(1/2+x,y,1/2+z)	2.54513
Cr1-Ti1(-1/2+x,y,1/2+z)	2.54513
Cr1-Ti1(1/2+x,y,-1/2+z)	2.54513
Cr1-Al1(x,1/2+y,1/2+z)	2.54513
Cr1-Al1(x,-1/2+y,1/2+z)	2.54513
Cr1-Al1(x,1/2+y,-1/2+z)	2.54513
Cr1-Al1(1/2+x,1/2+y,z)	2.54513
Cr1-Al1(-1/2+x,1/2+y,z)	2.54513
Cr1-Al1(1/2+x,-1/2+y,z)	2.54513
Cr1-Al1(1/2+x,y,1/2+z)	2.54513
Cr1-Al1(-1/2+x,y,1/2+z)	2.54513
Cr1-Al1(1/2+x,y,-1/2+z)	2.54513
Ti1-Ti1(x,1/2+y,1/2+z)	2.54513
Ti1-Ti1(x,-1/2+y,1/2+z)	2.54513
Ti1-Ti1(x,1/2+y,-1/2+z)	2.54513
Ti1-Ti1(1/2+x,1/2+y,z)	2.54513
Ti1-Ti1(-1/2+x,1/2+y,z)	2.54513
Ti1-Ti1(1/2+x,-1/2+y,z)	2.54513
Ti1-Ti1(1/2+x,y,1/2+z)	2.54513
Ti1-Ti1(-1/2+x,y,1/2+z)	2.54513
Ti1-Ti1(1/2+x,y,-1/2+z)	2.54513
Ti1-Al1(x,1/2+y,1/2+z)	2.54513
Ti1-Al1(x,-1/2+y,1/2+z)	2.54513
Ti1-Al1(x,1/2+y,-1/2+z)	2.54513

Distance

Bond Def.	Distance, Å
Ti1-Al1(1/2+x,1/2+y,z)	2.54513
Ti1-Al1(-1/2+x,1/2+y,z)	2.54513
Ti1-Al1(1/2+x,-1/2+y,z)	2.54513
Ti1-Al1(1/2+x,y,1/2+z)	2.54513
Ti1-Al1(-1/2+x,y,1/2+z)	2.54513
Ti1-Al1(1/2+x,y,-1/2+z)	2.54513
Al1-Al1(x,1/2+y,1/2+z)	2.54513
Al1-Al1(x,-1/2+y,1/2+z)	2.54513
Al1-Al1(x,1/2+y,-1/2+z)	2.54513
Al1-Al1(1/2+x,1/2+y,z)	2.54513
Al1-Al1(-1/2+x,1/2+y,z)	2.54513
Al1-Al1(1/2+x,-1/2+y,z)	2.54513
Al1-Al1(1/2+x,y,1/2+z)	2.54513
Al1-Al1(-1/2+x,y,1/2+z)	2.54513
Al1-Al1(1/2+x,y,-1/2+z)	2.54513